

DIY LAB IIT KHARAGPUR REPORT

SECURE ARDUINO - BASED DOOR LOCK WITH KEYPAD AND LCD DISPLAY

Problem statement : *Design and implement an Arduino-based door lock system that utilizes a keypad for user input and an LCD display for system feedback. The system should allow authorized users to enter a predefined code via the keypad to unlock the door, while denying access to unauthorized users*

INTRODUCTION :

In today's digital age, security is paramount, whether it's for our homes, offices, or personal belongings. Traditional lock and key mechanisms are being replaced by more sophisticated electronic systems that offer enhanced security and convenience. One such system is a door lock system based on Arduino, a popular open-source hardware and software platform widely used for building interactive electronic projects.

MOTIVATION :

- ❖ The use of a keypad adds an additional layer of security compared to traditional keys.
- ❖ Users can set custom PIN codes that are difficult to guess, reducing the risk of unauthorized access.
- ❖ Arduino provides a flexible platform for designing and implementing custom security features.
- ❖ Users can easily modify the code to incorporate advanced encryption algorithms, biometric authentication, or integration

with other smart home devices.

CHALLENGES FACED :

Security: Ensuring that the door lock system is secure from unauthorized access is paramount.

You'll need to implement robust encryption algorithms for communication between the keypad, Arduino,

and the locking mechanism. Additionally, you should incorporate mechanisms to prevent common attacks like

brute-force attacks on the keypad.

Power Consumption: Depending on the power source you're using for your Arduino-based system, power

consumption can be a concern, especially if the system needs to operate for extended periods without

recharging or replacing batteries. Optimizing code for efficiency and using low-power modes when the

system is idle can help mitigate this challenge.

Physical Security: The physical security of the hardware components, such as the Arduino board, keypad,

and locking mechanism, is crucial. Ensure that these components are housed in a secure enclosure to

prevent tampering or

unauthorized access.

COMPONENTS USED :

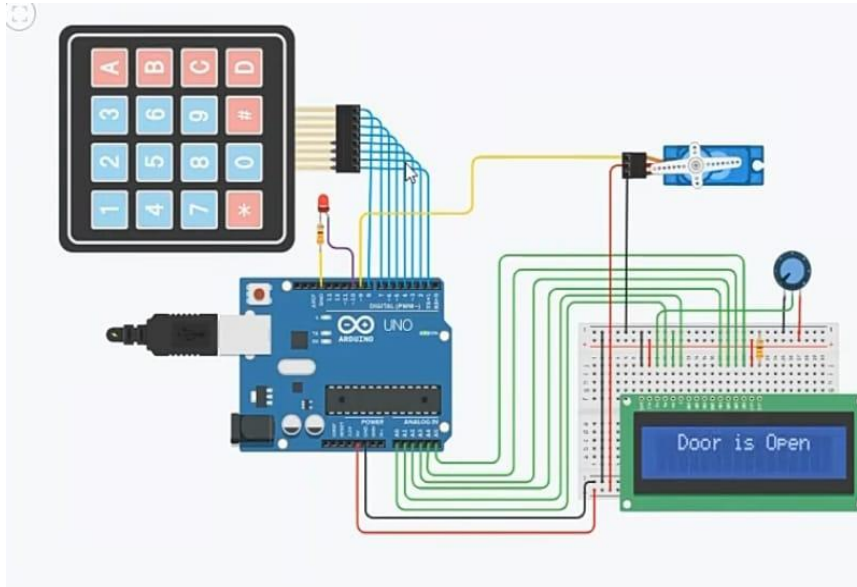
- Arduino Board
- Keypad
- LCD Display
- Electric Door Lock
- Power Supply
- Breadboard or Prototyping Board
- Jumper Wires
- Resistors
- Transistors/Relays

DETAILS OF THE SOLUTION :

- The secure Arduino-based door lock with keypad and display integrates a keypad for entering a secure code and a display for feedback.

- It utilizes cryptographic algorithms to securely store and verify access codes.
- The keypad entry triggers the Arduino to compare the input with stored codes, granting access upon successful verification.
- Additionally, the display provides real-time feedback, indicating whether access is granted or denied.
- This system ensures robust security while offering user-friendly functionality.

CIRCUIT DIAGRAM :



CODE OF THIS PROJECT :

```
#include <Servo.h>
#include <LiquidCrystal_I2C.h>
#include <Keypad.h>
#include <Password.h>

#define buzzer 13

Servo servo;
LiquidCrystal_I2C lcd(0x27, 16, 2);

char oldPassword[] = "1234"; // Old password, change as needed

String newPasswordString; // Hold the new password
char newPassword[7]; // Character array of newPasswordString
byte a = 5;
bool value = true;

Password password = Password(oldPassword); // Enter your password

byte maxPasswordLength = 6;
byte currentPasswordLength = 0;
const byte ROWS = 4; // Four rows
const byte COLS = 4; // Four columns char keys[ROWS][COLS] = {
  {'1', '2', '3', 'A'},
  {'4', '5', '6', 'B'},
```

```

{'7', '8', '9', 'C'},
{'*', '0', '#', 'D'}
};

```

```

byte rowPins[ROWS] = {10, 9, 8, 7};
byte colPins[COLS] = {6, 5, 4, 3};

```

```

Keypad keypad = Keypad( makeKeymap(keys), rowPins, colPins, ROWS, COLS );

```

```

void setup() {
  Serial.begin(9600);
  pinMode(buzzer, OUTPUT);
  servo.attach(12);
  servo.write(50);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(3, 0);
  lcd.print("WELCOME TO");
  lcd.setCursor(0, 1);
  lcd.print("DOOR LOCK SYSTEM");
  delay(3000);
  lcd.clear();
}

```

```

void loop() {
  lcd.setCursor(1, 0);
  lcd.print("ENTER PASSWORD");

```

```

  char key = keypad.getKey();
  if (key != NO_KEY) {
    delay(60);
    if (key == 'C') {
      resetPassword();

```

```

    } else if (key == 'D') {
      if (value == true) {
        doorlocked();
        value = false;
      } else if (value == false) {
        dooropen();
        value = true;
      }
    } else if (key == 'A') {
      changePassword();
    } else {
      processNumberKey(key);
    }
  }
}

```

```

void processNumberKey(char key) {
  lcd.setCursor(a, 1);
  lcd.print("*");
  a++;
  if (a == 11) {
    a = 5;
  }
  currentPasswordLength++;
  password.append(key);

  if (currentPasswordLength == maxPasswordLength) {
    doorlocked();
    dooropen();
  }
}

```

```

}

void dooropen() {
  if (password.evaluate()) { // Check with the old password
    digitalWrite(buzzer, HIGH);
    delay(300);
    digitalWrite(buzzer, LOW);
    servo.write(50);
    delay(100);
    lcd.setCursor(0, 0);

    lcd.print("CORRECT PASSWORD");
    lcd.setCursor(0, 1);
    lcd.print("OPEN THE DOOR...");
    delay(2000);
    lcd.clear();
    a = 5;
  } else {
    tone(buzzer, 1000);
    delay(200);
    noTone(buzzer);
    delay(200);
    tone(buzzer, 1000);
    delay(200);
    noTone(buzzer);
    delay(200);
    tone(buzzer, 1000);
    delay(200);
    noTone(buzzer);
    delay(200);
    lcd.setCursor(0, 0);
    lcd.print("WRONG PASSWORD!");
    lcd.setCursor(0, 1);
    lcd.print("PLEASE TRY AGAIN");
    delay(2000);
    lcd.clear();
    a = 5;
  }
  resetPassword();
}

void resetPassword() {
  password.reset();
  currentPasswordLength = 0;
  lcd.clear();
  a = 5;
}

void changePassword() {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Enter old PW:");

  String enteredPassword = "";
  while (true) {
    char key = keypad.getKey();
    if (key != NO_KEY && key != '#') {
      enteredPassword += key;
      lcd.print("*");
    } else if (key == '#') {
      lcd.clear();
      if (strcmp(enteredPassword.c_str(), oldPassword) == 0) { // Check if old password matches
        lcd.print("Enter new PW:");
        delay(1000);
        lcd.clear();
      }
    }
  }
}

```

```

String newEnteredPassword = "";
while (true) {
    char newKey = keypad.getKey();
    if (newKey != NO_KEY && newKey != '#') {
        newEnteredPassword += newKey;
        lcd.print("*");
    } else if (newKey == '#') {
        lcd.clear();

        strncpy(newPassword, newEnteredPassword.c_str(), sizeof(newPassword) - 1); // Update password
        newPassword[sizeof(newPassword) - 1] = '\0'; // Ensure null-termination
        password.set(newPassword);
        lcd.print("Password changed");
        delay(1000);
        lcd.clear();
        resetPassword();
        return;
    }
} else {
    lcd.print("Incorrect PW");
    delay(1000);
    lcd.clear();
    return;
}
}
}

void doorlocked() {
    if (password.evaluate()) { // Check with the old password
        tone(buzzer, 1000);
        delay(300);
        noTone(buzzer);
        servo.write(110);
        delay(100);
        lcd.setCursor(0, 0);
        lcd.print("CORRECT PASSWORD");
        lcd.setCursor(2, 1);
        lcd.print("DOOR LOCKED");
        delay(2000);
        lcd.clear();

        a = 5;
    } else {
        tone(buzzer, 1000);
        delay(200);
        noTone(buzzer);
        delay(200);
        tone(buzzer, 1000);
        delay(200);
        noTone(buzzer);
        delay(200);
        tone(buzzer, 1000);
        delay(200);
        noTone(buzzer);
        delay(200);
        lcd.setCursor(0, 0);
        lcd.print("WRONG PASSWORD!");
        lcd.setCursor(0, 1);
        lcd.print("PLEASE TRY AGAIN");
        delay(2000);
        lcd.clear();
        a = 5;
    }
}

```

```
resetPassword();  
}
```

LEARNING FROM THE PROJECT :

- A secure Arduino-based door lock with keypad and display offers several key benefits.
- Firstly, it provides enhanced security by requiring a personalized code for access, reducing the risk of unauthorized entry.
- Secondly, the integration of a display allows for user-friendly interaction, displaying prompts and feedback for smoother operation.
- Additionally, utilizing Arduino technology enables customization and flexibility, allowing users to tailor the lock system to their specific needs and preferences.
- Overall, this innovative solution combines security, usability, and adaptability for effective door access control.